

greens therom

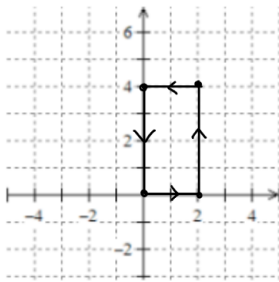
in simple words greens therom converts a walk around the edge of the shape into measurement of whats happen inside the shape

if the curve C is **Positively oriented** then greens therom allows you to convert a line integral around the boundary into a double integral over region inside

$$\int_C P dx + Q dy \longrightarrow \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Positively oriented means that its counterclockwise greens therom only works if its **Positively oriented**, this does not need to be conservative to apply greens therom, only needs to be **Positively oriented**

1. Use the Green's theorem to evaluate the following line integral along the positively oriented curve C a is the rectangle with vertices $(0,0)$, $(2,0)$, $(2,4)$, $(0,4)$



$$\int_C \underbrace{ye^x dx}_P + \underbrace{2e^x dy}_Q$$

$$\frac{\partial P}{\partial y} = e^x$$

$$0 \leq x \leq 2$$

$$0 \leq y \leq 4$$

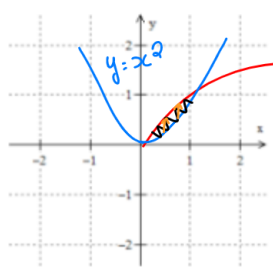
$$\frac{\partial Q}{\partial x} = 2e^x$$

$$\int_0^2 \int_0^4 (2e^x - e^x) dy dx \rightarrow \int_0^2 \int_0^4 e^x dy dx$$

$$\int_0^4 e^x dy \rightarrow \left[ye^x \right]_0^4 \rightarrow 4e^x$$

$$\int_0^2 4e^x dx \rightarrow \left[4e^x \right]_0^2 = \boxed{4e^2 - 1}$$

2. Use the Green's theorem to evaluate the following line integral along the positively oriented curve C bounded by the parabolas $y = x^2, x = y^2$



$$\int_C \underbrace{(3y + 5e^{\sqrt{x}})}_P dx + \underbrace{(10x + 9 \cos(y^2))}_Q dy$$

$$\frac{\partial Q}{\partial x} = 10$$

$$\frac{\partial P}{\partial y} = 3$$

$$x^2 \leq y \leq \sqrt{x}$$

$$0 \leq x \leq 1$$

We are going From bottom to top

$$\iint_C \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \rightarrow \int_0^1 \int_{x^2}^{\sqrt{x}} 7 dy dx$$

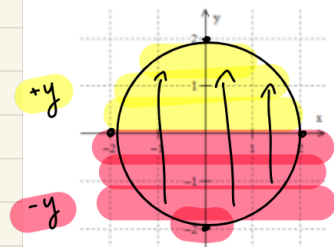
$$\left[7y \right]_{x^2}^{\sqrt{x}} \rightarrow 7x - 7\sqrt{x} \rightarrow \int_0^1 7x - 7\sqrt{x} dx$$

$$\left[\frac{7x^2}{2} - \frac{14}{3} x^{3/2} \right]_0^1 \rightarrow \frac{7}{2} - \frac{14}{3} = \boxed{-\frac{7}{6}}$$

3. Use the Green's theorem to evaluate the following line integral along the positively oriented curve

$$C: x^2 + y^2 = 4$$

$$\int_C \underbrace{2y^3 dx}_P - \underbrace{2x^2 dy}_Q$$



$$\frac{\partial P}{\partial y} = 6y$$

$$\frac{\partial Q}{\partial x} = -4x$$

$$-\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}$$

$$-2 \leq x \leq 2$$

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (-4x - 6y^3) dy dx$$

you can do it in polar coordinates as well

Polar

$$0 \leq \theta \leq 2\pi$$

$$0 \leq r \leq 2$$

$$\int_0^2 \int_0^{2\pi} (4r \cos \theta - 6(r \sin \theta)^3) d\theta dr$$

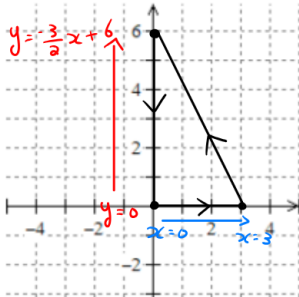
$$x = r \cos \theta$$

$$y = r \sin \theta$$

★ this is just a set-up

4. Use the Green's theorem to evaluate the following line integral along the positively oriented curve C : is the triangle from $(0,0)$ to $(0,4)$, to $(0,6)$ to $(0,0)$ where $\vec{F} = \underbrace{(y \cos x - xy \sin x)}_P, \underbrace{(xy + x \cos x)}_Q$

$$\int_C \vec{F} \cdot d\vec{r}$$



$$P(x,y) = y \cos x - xy \sin x$$

$$\frac{\partial P}{\partial y} = \cos x - x \sin x$$

$$Q(x,y) = xy + x \cos x$$

$$\frac{\partial Q}{\partial x} = y - x \sin x + \cos x$$

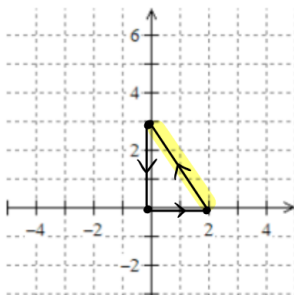
$$\int P dx + Q dy \rightarrow \iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \rightarrow \begin{matrix} 0 \leq y \leq -\frac{3}{2}x + 6 \\ 0 \leq x \leq 4 \end{matrix}$$

$$\int_0^4 \int_0^{-\frac{3}{2}x+6} (y - x \sin x + \cos x) - (\cos x - x \sin x) dy dx$$

$$\int_0^4 \int_0^{-\frac{3}{2}x+6} y dy dx$$

• this is Just setting
it up I will not solve
due to time and space

5. Use the Green's theorem to find the work done by the function $\vec{F} = \overset{P}{x(x+y)}\vec{i} + \overset{Q}{xy^2}\vec{j}$ in moving a particle from the origin along the x axis to (3,0) then along the line segment to (0,3), and then back to the origin along the y axis.



$$\iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \rightarrow \iint (y^2 - x) dA$$

$$\frac{\partial P}{\partial y} = x^2 + xy \rightarrow x$$

$$\frac{\partial Q}{\partial x} = xy^2 \rightarrow y^2$$

$$\begin{matrix} (0,3) & (3,0) \\ x_1, y_1 & x_2, y_2 \end{matrix}$$

$$M = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(0-3)}{(3-0)} = -1 \quad y = -x + 3$$

$$\int_0^3 \int_0^{-x+3} (y^2 + x) dy dx$$

• remember the Function comes First in the integral

$$0 \leq y \leq -x + 3$$

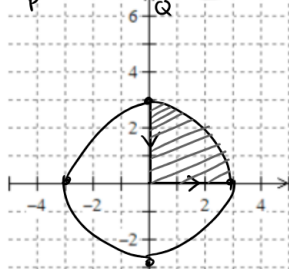
$$0 \leq x \leq 3$$

$$\int_0^3 \int_0^{-x+3} (y^2 + x) dy dx$$

This is just set up the integral is not solved

6. A particle starts at the origin along the x-axis to (3,0) then along the quarter circle $x^2 + y^2 = 9$ and down the **y-axis** to the origin. Use Green's theorem to find the work done by the force field $\vec{F}(x, y) =$

$$\underbrace{\sin x \vec{i}}_P + \underbrace{(\sin y + xy^2 + \frac{1}{2}x^2)}_Q \vec{j}$$



$$\frac{\partial P}{\partial y} = \sin x \Rightarrow 0$$

$$\frac{\partial Q}{\partial x} = \sin y + xy^2 + \frac{1}{2}x^2 \Rightarrow y^2 + x$$

$$0 \leq y \leq \sqrt{9-x^2}$$

$$0 \leq x \leq 3$$

$$\int_0^3 \int_0^{\sqrt{9-x^2}} (y^2 + x) dy dx$$

~~*~~ don't convert this to Polar coordinates

We are only setting this up Not solving

curl and divergence

a curl is a cross product:

You can either memorize this equation:

$$\text{curl } \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$$

(or)

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Clairaut theorem: if you get a curl of $\nabla \times \vec{F} = 0$ and it has a domain of all real numbers then you have a conservative vector field. So if you are asked if a field is conservative you can take the cross product of it and check its domain.

Divergence is a dot product

$$\text{div } \vec{F} = \nabla \cdot \vec{F}$$

examples:

Consider the following vector field $\vec{F} = xy^2z^2 \vec{i} + x^2yz^2 \vec{j} + x^2y^2z \vec{k}$

Find the curl ($\vec{F}$)

Find the div ($\vec{F}$)

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2z^2 & x^2yz^2 & x^2y^2z \end{vmatrix}$$

$$\vec{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz^2 & x^2y^2z \end{vmatrix} - \vec{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ xy^2z^2 & x^2y^2z \end{vmatrix} + \vec{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ xy^2z^2 & x^2yz^2 \end{vmatrix}$$

$$\vec{i} \left[\frac{\partial}{\partial y}(x^2y^2z) - \frac{\partial}{\partial z}(x^2yz^2) \right] - \vec{j} \left[\frac{\partial}{\partial x}(x^2y^2z) - \frac{\partial}{\partial z}(xy^2z^2) \right] + \vec{k} \left[\frac{\partial}{\partial x}(x^2yz^2) - \frac{\partial}{\partial y}(xy^2z^2) \right]$$

$$(2yz^2 - 2yzx^2)\vec{i} - (2xy^2z - 2zy^2x)\vec{j} + (2xyz^2 - 2yxz^2)\vec{k}$$

$$= 0$$

so you got zero and the domain of $\vec{F}$ is $\mathbb{R}^3$ which means you can plug in any number for the x, y, z variables so domain is all real numbers so this is also conservative

$$\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$$

$$\vec{F} = xy^2z^2\vec{i} + x^2yz^2\vec{j} + x^2y^2z\vec{k}$$

$$\text{div}(\vec{F}) = \frac{\partial}{\partial x}(xy^2z^2) + \frac{\partial}{\partial y}(-x^2yz^2) + \frac{\partial}{\partial z}(x^2y^2z)$$

$$\boxed{\text{div}(\vec{F}) = y^2z^2 - x^2z^2 + x^2y^2}$$

Surface Integrals

Surface integral of "f" over the surface "s" is given

$$\iint_s F(x,y,z) = \iint_s \vec{F}(\vec{r}(u,v)) |\vec{r}_u \times \vec{r}_v| dA$$

These two are Partial derivatives with respect to u and v

any surface with equation $z=g(x,y)$ can be regarded as a Parametric equation with Parametric equation with Parameters $x=x, y=y, z=g(x,y)$ and

$$\iint_s F(x,y) ds = \iint_s F(x,y,g(x,y)) \sqrt{\left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2 + 1} dA$$

if "s" is given by Vector Function $\vec{r}(u,v)$ then:

$$\iint_s \vec{F} \cdot d\vec{s} = \iint_s \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} ds \Rightarrow \iint_s \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) dA$$

and again the $\vec{r}$'s have Partial derivatives with respect to u and v

any surface with equation $z=g(x,y)$ can be regarded as a Parametric equation with Parameters $x=x, y=y, z=g(x,y)$

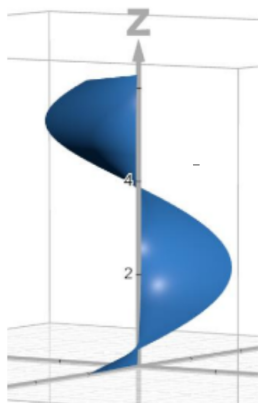
$$\iint_s \vec{F} \cdot d\vec{s} = \iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA \quad \text{Flux}$$

where $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$. This Formula assume upward orientation of s
For downward orientation we multiply by -1

1. Evaluate the surface integral $\iint_S y \, dS$ where S is a helicoid with vector equation $\vec{r}(u, v) = u \cos v \vec{i} +$

$$u \sin v \vec{j} + v \vec{k}, 0 \leq u \leq 2, 0 \leq v \leq \pi$$

~~$\vec{r}$~~ was provided so use first equation



$$F(x, y, z) = y$$

$$\iint_S y \cdot |\vec{r}_u \times \vec{r}_v|$$

$$\vec{r}(u, v) = u \cos v \vec{i} + u \sin v \vec{j} + v \vec{k}$$

$$\left. \begin{aligned} \frac{\partial \vec{r}}{\partial u} &= \cos v \vec{i} + \sin v \vec{j} + 0 \vec{k} \\ \frac{\partial \vec{r}}{\partial v} &= -u \sin v \vec{i} + u \cos v \vec{j} + 1 \vec{k} \end{aligned} \right\} \text{ apply cross product}$$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos v & \sin v & 0 \\ -u \sin v & -u \cos v & 1 \end{vmatrix}$$

$$\vec{i} \begin{vmatrix} \sin v & 0 \\ -u \sin v & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} \cos v & 0 \\ -u \sin v & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} \cos v & \sin v \\ -u \sin v & -u \cos v \end{vmatrix}$$

$$\vec{i} (\sin v) - \vec{j} (\cos v) + \vec{k} (u \cos^2 v + u \sin^2 v)$$

$$\sin v \vec{i} - \cos v \vec{j} + u \vec{k}$$

take magnitude (apply Pythagorean theorem)

$$\sqrt{(\sin v)^2 + (\cos v)^2 + (u)^2} \rightarrow \sqrt{1 + u^2}$$

$$\iint_S y \cdot \sqrt{1 + u^2} \, dA$$

$$\vec{r}(u, v) = \underbrace{u \cos v \vec{i}}_x + \underbrace{u \sin v \vec{j}}_y + \underbrace{v \vec{k}}_z$$

We will replace the y in the integral with $u \sin v$

$$\iint u \cos v \sqrt{1-u^2} \, dA$$

these were given:

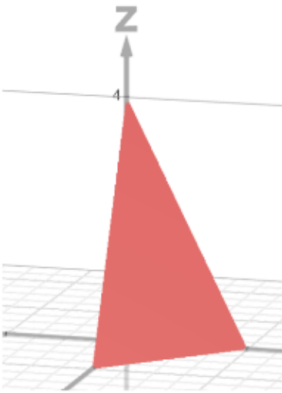
$$0 \leq u \leq 2$$

$$0 \leq v \leq \pi$$

$$\int_0^\pi \int_0^2 u \cos v \sqrt{1-u^2} \, du \, dv$$

★ this is just a set up
this won't be solved

2. Evaluate the surface integral $\iint_S xz \, dS$ where S is part of a plane $2x + 2y + z = 4$ that lies in the first octant.



You are given the equation $\iint_S xz \, ds$ You are given ds you will use the second equation

you can set parameters $x=x, y=y$ and $z=g(x,y)$
these parameters are what you are going to plug into equation

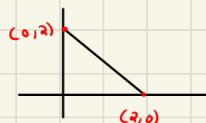
$$2x + 2y + z = 4 \Rightarrow z = \underbrace{4 - 2y - 2x}_{g(x,y)} \rightarrow ds = \sqrt{\left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2 + 1} \, dA \rightarrow ds = \sqrt{(2)^2 + (2)^2 + 1} = 3$$

$$\iint (x(4 - 2y - 2x)) \cdot 3 \, dA$$

If we set $z=0$

$$2x + 2y = 4$$

$$x + y = 2$$



• looking at graph the x is restricted from 0 to 2 so $0 \leq x \leq 2$

$$\begin{matrix} (0,2) & (2,0) \\ x_1, y_1 & x_2, y_2 \end{matrix}$$

$$M = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 2}{2 - 0} = -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -1(x - 0)$$

$$y = -x + 2$$

So "y" goes from

$$0 \text{ to } y = -x + 2$$

$$0 \leq x \leq 2$$

$$0 \leq y \leq -x + 2$$

$$\int_0^2 \int_0^{-x+2} (3x(4-2y-2x)) \, dy \, dx \rightarrow \int_0^2 \int_0^{-x+2} (12x - 6xy - 6x^2) \, dy \, dx$$

this is just a setup this won't be solved setting them up is more important